



RopeLink

KSA SA

Saudi Premier B2B Industrial Workforce & Rope Access Marketplace

Executive Leadership Brief

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RopeLink Platform: Executive Hardening Report & Technical Audit Specification

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Certified Status: **PILOT-READY — SECURITY VERIFIED** ✓

Strategic Objective: Transforming RopeLink into Saudi Arabia's premier B2B digital ecosystem for industrial manpower capacity exchange, monetizing idle turnaround crews (IRATA/NDT) across Jubail, Yanbu, and Vision 2030 megaprojects with hardened database-level governance.

AUTOMATED TESTS

62 / 62

100% Pass Rate

VULNERABILITIES

0

Clean Snyk & npm Overrides

PRODUCTION ENGINE

Turbopack

Next.js 16 + React 19

STORAGE & AUDIT RLS

Hardened

Isolated & Immutable

1 Completed Functional & Operational Pillars

Conditional Matchmaking Engine

Equipped with an upstream hard eligibility gate rejecting incompatible requests, followed by a normalized 4-factor scoring algorithm: **40% Specialty (IRATA/NDT)**, **30% Regional Proximity**, **20% Mobilization Timeline**, and **10% Headcount Sufficiency**.

Electronic Dual-Agreement Acceptance

In-app generation of turnaround subleasing agreements in SAR. Features tamper-proof dual electronic confirmation where agreements transition to active status only after explicit electronic acceptance by both contractor and supplier organizations.

SA Field Mobilization Tracker (7 Stages)

Enforces strict sequential progression (Agreement Signed → Roster Dispatched → Ajeer Permit → Gate Pass → HSE Induction → Active Execution → Turnaround Closeout) with database-level guards preventing milestone skipping.

2 Technical Hardening & Security Architecture

Enforced at PostgreSQL Engine

RBAC

Privilege Escalation & Self-Verification Guards:

Active PostgreSQL triggers prevent public signups from claiming admin roles and physically reject client attempts to self-verify or elevate permissions. Administrative provisioning is restricted to authorized server-side execution.

STORAGE

Storage RLS Isolation for Enterprise Documents:

The verification-docs bucket is marked strictly private with legacy open access policies eliminated. Dedicated RLS policies isolate documents per company (`auth.uid()/*`), preventing cross-tenant access to CR certificates.

AUDIT

Immutable Audit Trail & Governance Log:

Append-only audit logs table (`public.audit_logs`) denies direct client writes, populated exclusively by trusted database triggers upon agreement acceptance, mobilization advances, and administrative actions.

CHAT

Anti-Spoofing Real-Time Match Negotiation:

Chat sender IDs are cryptographically derived from the authenticated session, validated against match participant IDs via PostgreSQL RLS to eliminate sender forgery.

3 Operational Quality & Performance Infrastructure

Vercel Analytics & Speed Insights

⚡ Next.js 16 Turbopack

Upgraded to Next.js 16.3.4 with Turbopack, delivering sub-second page transitions and resolving all legacy Next.js CVEs.

📊 Real-Time Telemetry

Integrated `@vercel/analytics` and `@vercel/speed-insights` to monitor regional traffic patterns across Saudi Arabia.

🌐 Modular Architecture

Strict modularity enforced (all code files strictly < 200 lines) with complete bilingual synchronization across Arabic and English.

✓ Validated Pilot Release (Active Scope)

- B2B Manpower demand and available crew listings.
- IRATA/NDT specialty matching engine.
- Negotiation chat and dual electronic agreements.
- 7-stage mobilization tracker with Ajeer permit logging.
- Hardened RLS, anti-escalation triggers, and private storage.

🕒 Commercial Scaling (Future Phases)

- Daily field timesheets & 1.5x overtime calculations.
- Direct ZATCA Phase-2 electronic tax invoicing.
- Mada / Visa escrow payment gateways & collection tracking.
- Automated government API integration (MHRSD / Qiwa).

Executive Verdict:

The platform architecture is certified and hardened for controlled pilot deployment with foundational enterprise contractors in Jubail and Yanbu.

PILOT-READY
v0.3.2